Resilience Academy

SDI Training for ICT Professionals

CRD API in the Browser

Exploring Data Without Writing Code

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Day 4 | 30 Minutes Lecture | crd.resilienceacademy.ac.tz

Resilience Academy

What is an API?

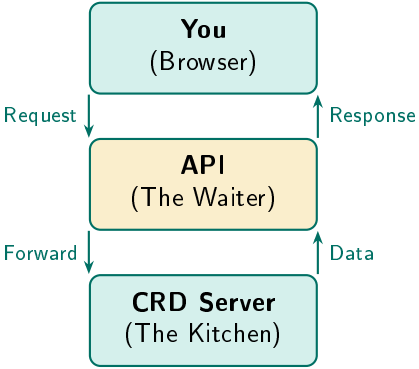
API in Simple Terms

API = Application Programming Interface

Think of it as a **waiter in a restaurant**:

- You (the customer) make a **request**: "I would like the chicken dish"
- The waiter takes your request to the **kitchen** (the server)
- The kitchen prepares the food and the waiter brings it back to you
- You never go into the kitchen yourself

An API is simply **a way for computers (or your browser) to talk to a server and get**



Good News

You have a website that

CRD Has a Built-In API

What is a REST API?

- REST = a standard way of organizing web addresses (URLs) so they return data
- CRD (GeoNode) comes with a REST API already built in — you do not install anything extra
- Every dataset, layer, and map on CRD can be accessed through a URL

The Base URL

The CRD API lives at:

`https://crd.resilienceacademy.ac.tz/api/v2/`

Everything starts from this address. You add to the end to request specific things:

- `/api/v2/datasets/` — list all

Try It Right Now!

Open your web browser (Chrome or Firefox) and visit:

`crd.resilienceacademy.ac.tz/api/v2/datasets/`

You will see a page full of structured text.

Exploring the API in Your Browser

What You See When You Visit the Datasets Endpoint

When you open `crd.resilienceacademy.ac.tz/api/v2/datasets/` in your browser, GeoNode shows a friendly interface with the raw data:

Key Information Returned

For each dataset, the API tells you:

- **title** — the name of the dataset
- **abstract** — description of what it contains
- **date** — when it was created or uploaded
- **category** — topic area (e.g., boundaries, health)
- **owner** — who uploaded it

Navigation

- The API returns results in **pages** (usually 10 per page)
- Look for `next` and `previous` links to browse pages
- The `total` field tells you how many datasets exist

JSON: The Language of APIs

What is JSON?

JSON = JavaScript Object Notation

It is simply **structured text that computers can read easily**. Think of it like a **form** with labels and values:

```
{
  "title": "Mwanza Businesses",
  "owner": "admin",
  "date": "2026-03-07",
  "category": "Economy",
  "total_features": 342,
  "is_published": true
}
```

Reading JSON — It is Easy!

- **Curly braces** `{ }` wrap each object (like a record)
- **Square brackets** `[ ]` wrap lists of items
- Each piece of data has a **label** (left of the colon) and a **value** (right of the colon)
- Text values are in **quotes**: `"Mwanza"`
- Numbers have **no quotes**: 342
- True/false values: `true`, `false`

You Do Not Need to Write JSON

You only need to read it. The API generates

Searching for Data with the API

Adding Search Parameters to the URL

You can filter API results by adding `?search=` to the URL. This is like using the search box on the CRD website, but directly in the address bar.

Try These URLs in Your Browser

- **Search for Mwanza data:**

  `.../api/v2/datasets/?search=mwanza`

- **Search for flood data:**

  `.../api/v2/datasets/?search=flood`

- **Search for education facilities:**

  `.../api/v2/datasets/?search=education`

- **Filter by category:**

How URL Parameters Work

- ? starts the parameters
- `search=mwanza` is the filter
- & adds more filters
- `page_size=5` limits results to 5

These are just text you type in the browser address bar — no programming required.

Postman: A Visual Tool for APIs

What is Postman?

- A **free application** (download from `postman.com`)
- Provides a **graphical interface** for working with APIs
- No coding — you type a URL, click Send, and see results
- Formats JSON responses with colours and folding
- Saves your favourite requests for reuse

Why Use Postman Instead of the Browser?

- **Better formatting** — JSON is colour-coded and collapsible
- **History** — keeps a record of all requests you have made
- **Collections** — save and organise your API calls
- **Headers and Auth** — easily add login credentials when needed
- **Export** — share your API calls with colleagues

Using Postman with CRD

When Do APIs Matter? + Session Summary

Real-World Uses of the CRD API

1. **Automated reports**
   A dashboard pulls data from CRD every morning and generates a summary
2. **Mobile applications**
   A phone app shows the nearest health facility using CRD data
3. **Website integration**
   Embed a live map on your organisation's website
4. **Data synchronisation**
   Automatically update a national database when new data is uploaded to CRD

You Do Not Need to Code

As an ICT professional, your role is to:

- **Understand** that the API exists
- **Know** what it can do
- **Communicate** with developers about what data is available
- **Test** API calls in the browser or Postman

The actual programming is done by software developers — but **you** are the one who knows what data exists and how it is structured.